



ILOCOS SUR POLYTECHNIC STATE COLLEGE
Tagudin Campus, Tagudin, Ilocos Sur

**NEXHRIS: A SMART AND NEXT-GENERATION HUMAN RESOURCE
INFORMATION SYSTEM FOR ILOCOS SUR POLYTECHNIC STATE
COLLEGE**

EIDREF JAKE S. MANALANSAN

SHERAN S. LEAL

ARVIE KEITH M. MANZANO

DIANA SOFIA L. STA. ANA

YURI BIB P. SOMERA

ILOCOS SUR POLYTECHNIC STATE COLLEGE
COLLEGE OF ARTS AND SCIENCES
TAGUDIN, ILOCOS SUR

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

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Chapter 1

INTRODUCTION

Background of the Study

The rapid advancement of technology has fundamentally transformed the way organizations manage their human resources. Across the globe, institutions are shifting away from manual, paper-based administrative processes toward integrated digital platforms that offer efficiency, accuracy, and accessibility. In government higher education institutions, Human Resource (HR) offices are primarily responsible for maintaining, organizing, and safeguarding the employment records of all personnel. These records, collectively known as the 201 files, contain the complete employment history of an employee, including personal information, appointment papers, service records, certificates of eligibility, and other official employment documents. The proper management and accessibility of these records are critical to the effective administration of human resources within any institution.

Internationally, several studies confirm the urgent need for digital employee document management systems in organizations. Wahdaniah et al. (2023) demonstrated that human resource management transformation in the digital age results in more organized recordkeeping, reduced document retrieval time, and improved overall HR operational efficiency across diverse institutional settings. Similarly, Abaci and Medeni (2022) conducted a case study on the efficiency of electronic document management systems and established that digitizing employee records reduces retrieval time, minimizes physical storage requirements, and enhances organizational transparency. Orioque et al. (2024) further affirmed that an online contextualized document management system aligned with the operational



environment of an organization significantly enhances information access and reduces misplaced documentation. Dres and Rabut (2025) developed eDOCS, a web-based file management system, and found it to be highly effective in improving document organization, accessibility, and retrieval speed compared with traditional manual filing methods. Furthermore, Panfilova et al. (2024) noted that the integration of automated document management systems into institutional workflows facilitates diagnostic monitoring, reduces human error, and improves documentation lifecycle management, all of which are directly relevant to the employee file management needs of human resource offices in academic institutions.

In the Philippines, the digitalization of employee and institutional records has similarly gained traction among higher education institutions. Purcia and Velarde (2022) conducted a study on student registration and records management in private universities and found that the majority of institutions still rely on paper-based processes, highlighting a persistent gap in digital adoption and the urgent need for electronic records systems. Pagayonan (2022) developed an Electronic Document Management System (eDMS) for a higher education institution and found that the system improved the organization, storage, and retrieval of institutional records while reducing dependency on physical filing systems. Encarnacion et al. (2024) developed a web-based document management system for an educational office in Pagadian City that streamlined the dissemination, submission, and collection of pertinent documents, addressing challenges typically experienced by office employees in managing paper-based records. More recently, Rabulan and Glina (2025) developed a web-based file management system for the Office of the National Service Training Program at Sorsogon State University, which effectively centralized student information and



office records, resulting in a more secure, accurate, and faster file management process that directly reduced the burden on office personnel.

At the regional and provincial level, the need for digital records management has been consistently identified, particularly at ISPSC Tagudin Campus itself. Angala et al. (2023) developed and implemented a web-based Document Management System (DMS) at ISPSC Tagudin Campus and found that respondents were generally very satisfied with how the system was used to arrange, store, track, and manage documents within the campus, affirming that digital document systems are well received at the institution. Bulong et al. (2025) similarly focused on ISPSC Tagudin Campus and developed DocuTrack, a web-based document management system featuring role-based access control, cloud storage integration, and an activity log, confirming that paper-based document management remains a critical challenge for the institution as it continues to grow in terms of personnel, programs, and administrative responsibilities. Beyond ISPSC, Sebucao-Orias (2023) examined the records management practices of a state university in the Philippines and identified significant deficiencies in documentation processes, while Taguam (2024) established that non-compliance with proper records management protocols in state universities leads to operational inefficiencies, both underscoring the need for structured digital records management systems in similar institutions. The Ilocos Regional Development Plan 2023–2028 further identifies the enhancement of institutional efficiency and the adoption of digital governance tools as priority directions for all government agencies and institutions in the region (Regional Development Council, 2023). The Data Privacy Act of 2012 (Republic Act No. 10173) likewise mandates that all government institutions, including state colleges in Ilocos Sur, ensure the confidentiality, integrity,



and availability of the personal data in their custody, which directly necessitates the adoption of secure digital systems for managing sensitive employee records. Furthermore, the national directive on digital transformation articulated in the Philippine Development Plan 2023–2028 reinforces the importance of transitioning from paper-based to digital HR records management in institutions such as ISPSC (Department of Information and Communications Technology, 2023).

The implementation of NexHRIS at Ilocos Sur Polytechnic State College, Tagudin Campus is particularly justified by the current state of HR document management in the institution. The ISPSC HR Office maintains all employee 201 files on paper and in Excel workbooks, keeping a Personal Data Sheet per employee, a separate form per leave application, and two handwritten ledger cards per person. Leave approval requires a printed form to be physically routed between three offices for signature. These manual systems are prone to document loss, physical deterioration, unauthorized access, and delays in document retrieval. As the institution continues to grow in terms of personnel and administrative responsibilities, the volume of employee records that must be managed likewise increases, making traditional paper-based processes increasingly inefficient and difficult to sustain. The absence of a centralized digital repository for employee files requires HR personnel to spend considerable time locating, reproducing, and verifying documents that could otherwise be accessed instantly through a secure electronic system. Employees, on their part, must still visit the HR Office to update their Personal Data Sheet, to check their leave credits, and to file a leave application, adding to the workload of HR personnel who have to retrieve records and compute balances manually for every request.



A review of the systems cited above reveals a consistent limitation: all of them are document management systems, not human resource systems. The eDMS of Pagayonan (2022), eDOCS of Dres and Rabut (2025), the contextualized system of Orioque et al. (2024), the web-based system of Encarnacion et al. (2024), and the file management system of Rabulan and Glina (2025) are each designed to store, categorize, track, and retrieve files. The same is true of the two systems developed for ISPSC Tagudin Campus itself: the DMS of Angala et al. (2023) and DocuTrack by Bulong et al. (2025) both centralize institutional documents, and DocuTrack adds role-based access and activity logging, but both stop at document handling. In every one of these systems, a document is treated as a file to be uploaded and retrieved, and none of them processes the transaction that the document represents.

NexHRIS differs from all of them in four specific ways. First, none of the reviewed systems handles the Personal Data Sheet as a governed submission with a review-and-return cycle and retained versions; each would treat a PDS as one more uploaded file. Second, none of them routes a document through a multi-level approval chain in which the route is determined by the rank of the applicant, nor records which approvals were skipped and why. Third, none of them maintains a running-balance record such as the leave and service credit ledger cards of the campus, where correcting one entry must recompute every balance below it. Fourth, none of them is concerned with preserving the exact printed format of an official government form, because a document management system stores the file that the user uploaded and never has to regenerate it. These are precisely the day-to-day HR transactions that ISPSC employees still complete manually, even though the campus already has a working document management system. The gap, therefore, is not the digitization of storage, which the



literature has addressed thoroughly, but the digitization of the HR transactions themselves, which NexHRIS is designed to fill.

Recognizing this gap, NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College was conceptualized to go beyond the storage-and-retrieval model demonstrated by existing systems. Rather than covering the entire employee 201 file, the study focuses specifically on the components with which employees interact most directly and repeatedly, the Personal Data Sheet, leave records, personal information, and HR policies and guidelines, allowing employees to take part in managing their own HR transactions and to stay informed of institutional policies, instead of relying solely on manual, HR-mediated processes. Through the development of NexHRIS, the study seeks to address the limitations of existing manual practices, strengthen records security and accessibility, improve operational efficiency, and support compliance with government policies on digital transformation and data privacy. Ultimately, the study aims to contribute to the modernization of human resource management practices in ISPSC and to provide a model for the implementation of HR-focused information systems in higher education institutions.

Conceptual Framework of the Study

The conceptual framework uses the Input–Process–Output (IPO) model. The Input includes employee 201 files, particularly PDS and leave records, employees’ personal information, and HR policies and guidelines. The Process involves the design, development, and implementation of NexHRIS to manage employee information, leave applications and records, and HR policies through a centralized digital platform. The Output consists of centralized employee records, online PDS and leave



management, digital employee identification records, and accessible HR policies and guidelines. Overall, the IPO model shows how NexHRIS transforms manual HR records into an integrated digital system that improves record management and HR services at Ilocos Sur Polytechnic State College.

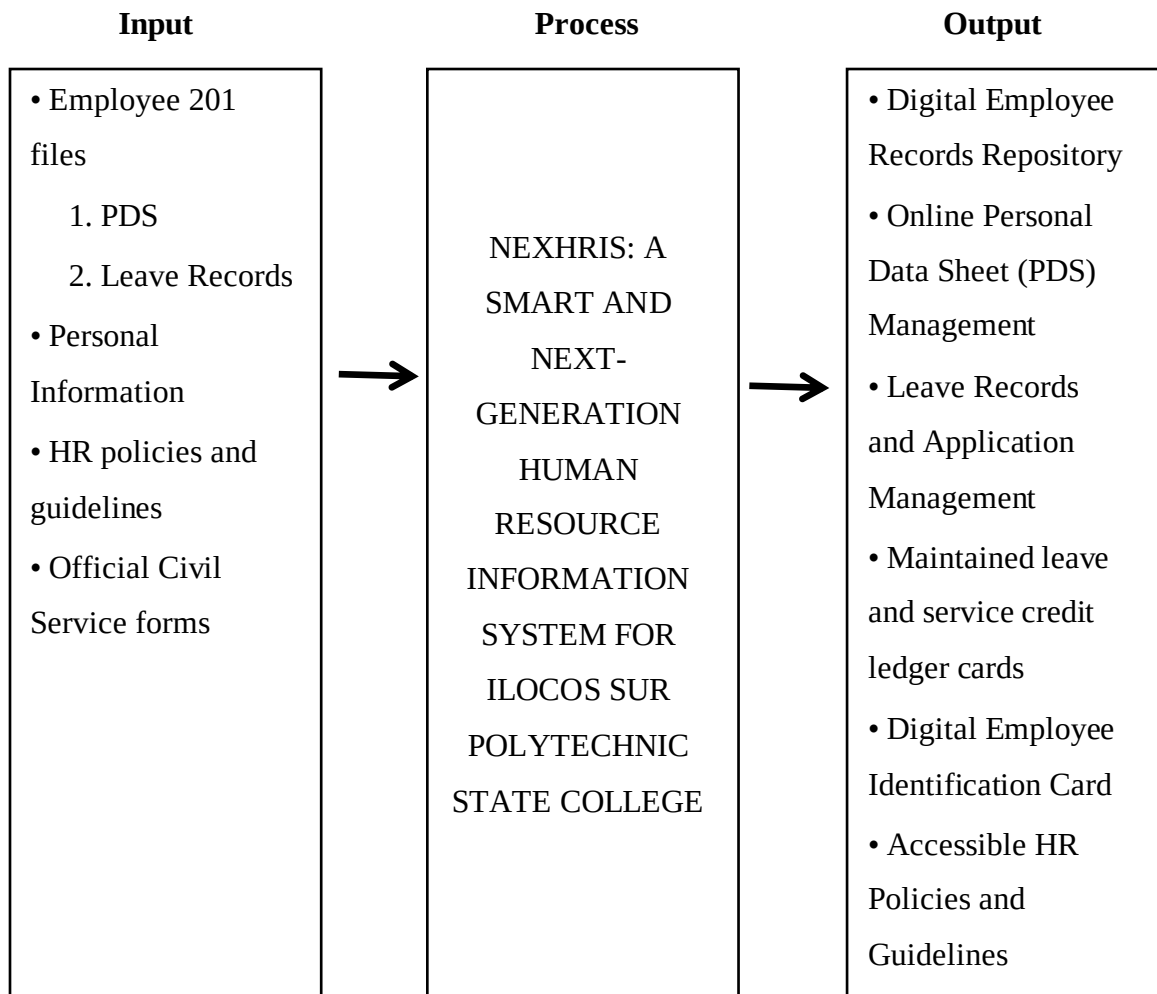


Figure 1. Research Paradigm

Objectives of the Study

The main objective of this study is to develop NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College to support the secure and efficient digital management of employee Personal



Data Sheets (PDS) and leave records at ISPSC Tagudin Campus. Specifically, this study aims:

1. To analyze the current employee file management processes of the ISPSC HR Office to serve as the basis for the system design and development;
2. To determine the system requirements for the development of NexHRIS, a web-based Human Resource Information System for the secure management of employee Personal Data Sheets, leave records, and personal information at ISPSC Tagudin Campus;
3. To evaluate the level of acceptability of NexHRIS using the Technology Acceptance Model (TAM); and
4. To determine whether there is a significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use.

Hypothesis

Ho: There is no significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use.

Scope and Limitation of the Study

The study covers the design, development, and implementation of NexHRIS for ISPSC Tagudin Campus, focusing on four components of the employee 201 file: the Personal Data Sheet (PDS), leave records, personal information, and HR policies. The HR Administrator publishes numbered official blank forms, including the PDS (CS Form No. 212), Application for Leave (CS Form No. 6), and leave ledger card, which



employees download, complete, and upload for review. The HR Administrator may approve or return PDS submissions with remarks, while submitted versions are retained. Leave applications follow a rank-based approval chain involving the Dean, HR Administrator, and Campus Director, depending on the applicant's position, after which approved leave is recorded in the appropriate ledger. The system supports four roles, HR Administrator, Campus Director, Dean, and Employee with query-level access restrictions, digital identification cards with QR verification, HR policy acknowledgment, announcements and notifications, role-based dashboards, a leave calendar, printable reports, and audit trails. It also requires acceptance of the Terms and Conditions and provides additional email-based six-digit verification for administrative accounts. NexHRIS is web-based and accessible through internet-connected devices.

The study is limited to selected components of the employee 201 file and does not digitize appointment papers, service records, performance ratings, or other remaining documents. It does not perform OCR, automatically extract or verify information from uploaded forms, or validate their authenticity. Leave applications are limited to vacation and sick leave, while other leave types remain manually processed. Leave credits and ledger balances are maintained by the HR Administrator, and the system does not calculate credits from attendance or biometric data or automatically determine the appropriate ledger. Official forms must be completed using Microsoft Excel or LibreOffice Calc, while documents are converted to PDF through server-side LibreOffice or a built-in renderer when necessary. The policies module is limited to reading and acknowledgment, and the digital ID has no RFID functionality. NexHRIS does not integrate with external government systems such as the CSC, GSIS, or DBM,



supports only ISPSC Tagudin Campus, has no dedicated mobile application or offline functionality, and requires an internet connection.

Importance of the Study

The development of NexHRIS is expected to provide significant benefits to various stakeholders by improving the management, accessibility, and security of employee records. The beneficiaries of this study are discussed as follows:

To the HR Office of ISPSC Tagudin Campus. NexHRIS provides a centralized and secure platform for managing employee 201 files and personnel records. It reduces the time and effort needed to store, retrieve, and verify documents while streamlining leave applications and ledger maintenance.

To the Employees of ISPSC Tagudin Campus. The system provides an accessible platform for submitting and printing their PDS, filing and monitoring leave applications, viewing ledger cards, and accessing personal information, reducing the need for routine visits to the HR Office.

To the Deans and Campus Director. NexHRIS centralizes leave applications awaiting their decision and provides access to relevant leave information, making the approval process more convenient and organized.

To Ilocos Sur Polytechnic State College (ISPSC). The institution benefits from improved records security, reduced risk of document loss or damage, and more organized employee records management. The system also supports compliance with the Data Privacy Act of 2012 and institutional digital transformation initiatives.



To the Researchers. The study provides practical experience in designing, developing, and implementing a web-based system while enhancing their skills in system development, document processing, and evaluation.

To Future Researchers. The study may serve as a reference for developing similar HR information systems and employee records management solutions and may support further research and innovation in HR technology.



Chapter 2

METHODOLOGY

This chapter presents the methodology to be employed in the development of NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College. It discusses the research design, software development model, project timeline, project assignments, target population and locale, research instruments, and the methods of data analysis to be used in the study.

Research Design

This study used a descriptive developmental research design. The descriptive component involved gathering data on the employee file management practices and challenges of the ISPSC HR Office through interviews and observations, which served as the basis for identifying the required system features and design specifications. The developmental component focused on the actual design, development, and implementation of NexHRIS as a functional web-based human resource information system tailored to the needs of the HR Office and the employees of ISPSC Tagudin Campus.

According to Kosie and Lew-Williams (2022), developmental research is the systematic process of designing, developing, and evaluating programs or systems to ensure internal consistency and effectiveness. The descriptive component documented each stage of development, while the developmental component emphasized iterative enhancement and continuous refinement based on user needs and the results of system testing. This design guided the entire research process, from requirements gathering through system development, testing, and final evaluation.

In the context of this study, the descriptive developmental design began with an analysis of the current HR document management process at ISPSC Tagudin Campus, followed by the identification of the specific features required in NexHRIS based on the actual needs of the HR Administrator and the employees. The system was then designed, developed, and iteratively tested, with refinements made based on user feedback, until the final version was deployed and formally evaluated.

Software Model

This study adopted the Rapid Application Development (RAD) model as its software development methodology. RAD emphasizes rapid prototyping, iterative development, and continuous user feedback, making it suitable for institutional systems with defined requirements and modular development (Pressman & Maxim, 2020). In this study, RAD supported the development of NexHRIS by allowing employee accounts, PDS management, leave applications and approvals, ledger maintenance, and HR policies to be developed and tested in stages, with feedback from the HR Administrator incorporated throughout the process.

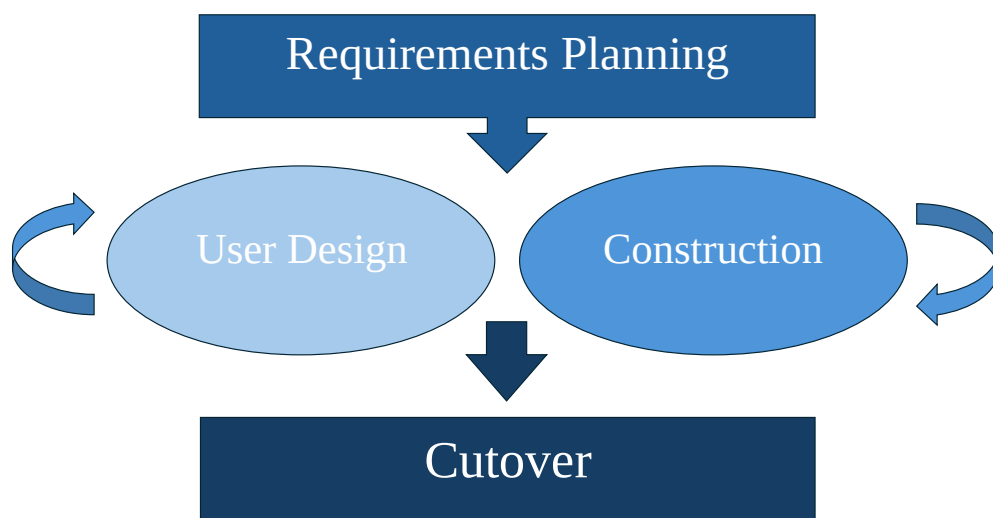


Figure 2. Rapid Application Development



Requirements Planning. The researchers conducted structured interviews and consultations with the ISPSC HR Administrator to identify challenges in employee file management and define the functional requirements of NexHRIS. These included a centralized repository for Personal Data Sheets and leave records, secure role-based access, multi-level leave filing and monitoring, maintenance of leave and service credit ledger cards, and faithful reproduction of the campus's official Civil Service forms.

User Design. A user-centered approach was used to design the interfaces of all modules. Feedback from the HR Administrator guided the design of the PDS review screens, leave review queue, ledger maintenance, and employee account management, while employee experience informed the PDS workspace, leave filing and tracking, and profile and identification card screens. Each role was provided with a navigation menu displaying only authorized modules. The database was designed as a relational schema, and the system design was documented through use case diagrams, data flow diagrams, an entity relationship diagram, and a flowchart.

Construction. NexHRIS was developed using Laravel 12 and PHP for the back end, Blade with Tailwind CSS and Alpine.js for the interface, Chart.js for dashboard charts, and MariaDB for data storage. PhpSpreadsheet validates uploaded Civil Service workbooks, while LibreOffice converts them to PDF, with PhpSpreadsheet and Dompdf providing a fallback renderer. Dompdf also generates approval sheets, ledger cards, and management reports. Simple QrCode generates digital ID QR codes, while Symfony HTML Sanitizer cleans HR policy content. Modules were developed sequentially, tested, refined, and integrated. An automated test suite was executed after each change and ultimately comprised 530 tests covering authentication, role boundaries, approval workflows, ledger calculations, document conversion, policy



sanitization, and personnel-record confidentiality, all of which passed. Browser-based testing was also conducted across all four roles on desktop and mobile screen sizes.

Cutover. The completed system was deployed on a Hostinger KVM 1 VPS with 1 vCPU, 4 GB RAM, and 50 GB storage, running Ubuntu 24.04 LTS, Nginx, PHP 8.3, MariaDB, and LibreOffice, with HTTPS secured through Let's Encrypt. Development was conducted locally using PHP 8.2 and MariaDB. Personnel records are stored securely outside the web-accessible directory and accessed only through the application after role verification. The phase also included user training, final adjustments based on feedback, and administration of the TAM questionnaire.

Project Plan

The development of NexHRIS will follow the four phases of the Rapid Application Development (RAD) Model. The project is planned across a period covering June 2026 to September 2026, with each phase assigned specific milestones and deliverables to ensure timely and organized system development.

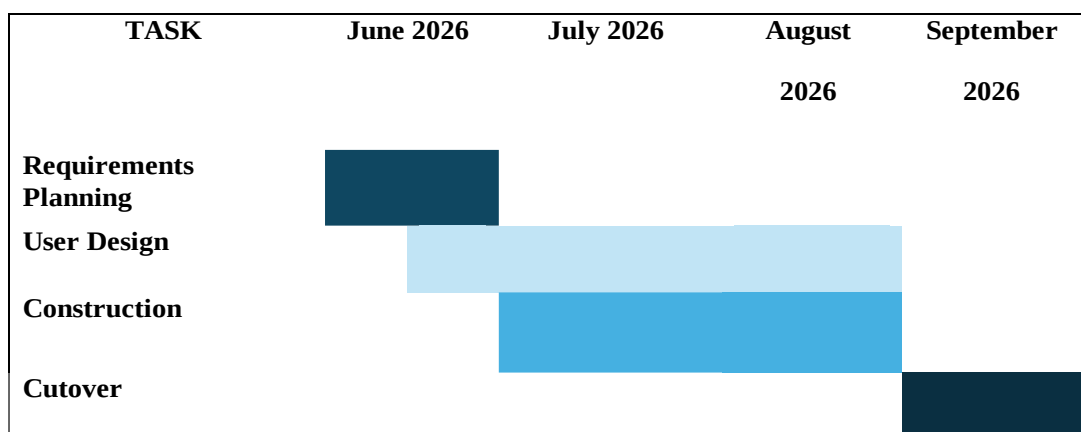


Figure 3: Project Schedule



Project Assignment

The roles and responsibilities of the project team for NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College are presented in Table 1. Each member is assigned specific tasks based on their skills and areas of specialization to ensure proper development and completion of the system.

Table 1. Project Assignment

Roles	Name	Function
Project Manager	Eidref Jake S. Manalansan	• Responsible for coordinating the project team and maintaining open communication among all members.
System Analyst and Designer	Eidref Jake S. Manalansan Diana Sofia L. Sta. Ana	• Coordinate the technical team in resolving challenges and implementing practical solutions.
Programmer and Developer	Eidref Jake S. Manalansan	• Manage the systems planning design team and built the project.
QA/ Tester	Diana Sofia L. Sta. Ana Sheran S. Leal	• Responsible for checking debugging issues, evaluating software needs, and communicating findings to the project team.
Documenter/ Technical Writer	Eidref Jake S. Manalansan Diana Sofia L. Sta. Ana Sheran S. Leal Yuri Bib P. Somera Arvie Keith M. Manzano	• Publish documents to relevant parties and report the overall project status.



Population and Locale of the Study

The study will be conducted at Ilocos Sur Polytechnic State College (ISPSC), Tagudin Campus, located in Tagudin, Ilocos Sur. This institution serves as the locale of the study as NexHRIS is specifically developed to address the employee file management needs of the ISPSC HR Office.

The respondents of this study will consist of two groups. The first group is composed of the two (2) authorized users of the NexHRIS system within the HR Office, the HR Administrator and the HR Assistant, both of whom share the same level of system access and are the primary operators of the system. Since the total number of HR Office users is only two (2), total enumeration will be applied for this group.

The second group consists of the teaching and non-teaching personnel of ISPSC Tagudin Campus with a total population of 142 employees. The researchers used Slovin's Formula to determine the appropriate sample size for this group. From a population size of 142, a sample size of 105 employees was chosen to represent the population with a 5% margin of error.

Slovin's Formula

$$n = \frac{N}{1+N(e^2)}$$

Whereas:

n = no. of samples

N = total of population

e = margin of error

Given:

N = 142

e = 0.05(5%)



Slovin's Formula Calculation:

$$n = \frac{142}{1+142(0.05)^2} = n = 104.80 \approx \mathbf{105}$$

The 105 employee-respondents will be selected through simple random sampling, wherein each of the 142 employees will be given an equal chance of being selected until the required sample size is reached.

Table 2: Distribution of All Respondents

Respondents	Sample Size
HR Administrator	1
HR Assistant	1
Employees	105
TOTAL	107

Table 2 presents the distribution of all respondents. This study will involve a total of 107 respondents, composed of two (2) HR Office personnel, the HR Administrator and HR Assistant who will serve as the primary system administrators, and 105 teaching and non-teaching personnel of ISPSC Tagudin Campus who will serve as employee-users of NexHRIS. The respondents were identified based on their direct involvement with or as intended users of the NexHRIS system, ensuring that both the administrative and employee perspectives are represented in the system evaluation.

Research Instruments

The data for this study will be collected through two methods. First, structured interviews will be conducted with the ISPSC HR Administrator prior to system



development to determine the current employee file management processes, challenges, and requirements that will form the basis for the NexHRIS feature design. Second, after system deployment, the Technology Acceptance Model (TAM) questionnaire will be administered to all respondents, the HR Administrator, HR Assistant, and selected employee-users to evaluate the acceptability of the system.

The Technology Acceptance Model (TAM), originally developed by Davis (1989), is a widely validated and widely used theoretical framework for evaluating user acceptance of information systems. TAM posits that two primary constructs, Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) to determine a user's attitude toward using a system and, ultimately, their behavioral intention to use it. Perceived Usefulness refers to the degree to which a user believes that using the system will enhance their job performance, while Perceived Ease of Use refers to the degree to which a user believes that the system will be free of effort. The TAM questionnaire to be used in this study will assess both constructs using a five-point Likert scale, with 1 indicating strong disagreement and 5 indicating strong agreement with each statement.

The TAM instrument is particularly appropriate for this study because NexHRIS is a new system being introduced to users who have no prior experience with a digital employee file management system. Evaluating the system through TAM will allow the researchers to determine whether the users find NexHRIS useful for their HR document management tasks and whether they find it easy to operate, which are the two most critical determinants of system adoption and continued use in institutional settings.



Data Analysis

The researchers will use appropriate statistical and qualitative tools to analyze the data gathered for each objective of the study. The following presents the data analysis methods to be applied per objective.

For Objective 1, which aims to analyze the current employee file management processes of the ISPSC HR Office, the qualitative data gathered through structured pre-development interviews with the HR Administrator and HR Assistant will be analyzed using thematic analysis. This involves systematically organizing and categorizing the responses into recurring themes related to the current HR document management workflow, the challenges and inefficiencies experienced by the HR Office, and the specific system features and requirements identified by the HR personnel. The themes derived from this analysis will directly serve as the basis for the system design, feature prioritization, and functional requirements of NexHRIS.

For Objective 2, which aimed to determine the system requirements for the development of NexHRIS, the functional and non-functional requirements identified during requirements planning were documented and organized into system specifications. These requirements were validated with the HR Administrator and translated into the modules, user roles, and workflows of the system, ensuring that the developed system directly addresses the needs identified in Objective 1.

For Objective 3, which aims to evaluate the level of acceptability of NexHRIS based on the responses of the HR Administrator and the employees of ISPSC Tagudin Campus using the Technology Acceptance Model (TAM), the researchers will use mean and standard deviation to analyze the quantitative data gathered from the TAM



questionnaire. These descriptive statistical tools will be used to determine the level of acceptability of NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College as perceived by the HR Administrator, HR Assistant, and employee-users of ISPSC Tagudin Campus in terms of Perceived Usefulness (PU) and Perceived Ease of Use (PEOU).

Each item in the TAM questionnaire will be rated by the respondents using a five-point Likert scale as presented in Table 3.

Table 3. Likert Scale for the TAM Evaluation of NexHRIS

Scale	Verbal Interpretation
5	Strongly Agree
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

The computed mean scores for each TAM construct, particularly Perceived Usefulness and Perceived Ease of Use, will be interpreted using the mean range scale presented in Table 4. This interpretation will help determine the respondents' level of agreement with each construct and assess the overall acceptability of the proposed system.



Table 4. Mean Range Interpretation for TAM Evaluation of NexHRIS

Mean Range	Interpretation
4.21 – 5.00	Strongly Agree
3.41 – 4.20	Agree
2.61 – 3.40	Neutral
1.81 – 2.60	Disagree
1.00 – 1.80	Strongly Disagree

For Objective 4, which aims to determine whether there is a significant difference between the existing manual employee file management process and the developed NexHRIS system in terms of perceived usefulness and perceived ease of use, the researchers will use the Wilcoxon Signed-Rank Test. This non-parametric statistical test is appropriate for comparing two related conditions: the manual process and the NexHRIS system using the same group of respondents, particularly when the data is ordinal in nature, as in the case of Likert-scale responses. The respondents will be asked to rate both the current manual process and the NexHRIS system using the same TAM-based items on Perceived Usefulness and Perceived Ease of Use. The Wilcoxon Signed-Rank Test will then be applied to determine whether a statistically significant difference exists between the two sets of ratings. A significance level of 0.05 will be used as the decision threshold. If the computed p-value is less than 0.05, the null hypothesis that there is no significant difference between the manual process and NexHRIS will be rejected, indicating that a significant difference exists in favor of the developed system.



Chapter 3

RESULTS AND DISCUSSION

This chapter presents the findings of the study in the order of its objectives: the current employee records management practices of the institution, the system requirements of NexHRIS and the features of the developed system, the acceptability of the system, and the comparison between the manual process and NexHRIS.

Findings

The following findings were identified from the interview and consultation with the HR Administrator regarding the existing employee records and HR data management processes. These findings provided the basis for identifying the current challenges, determining the system requirements, and developing NexHRIS to address the needs of the ISPSC Tagudin Campus HR Office.

The Current Process of Employee Records and HR Data Handling

The HR process of ISPSC Tagudin Campus relied primarily on manual, paper-based, and spreadsheet-based procedures. Employee 201 files, including the Personal Data Sheet, leave records, and other personnel documents, were physically stored and organized by the HR Office. Each employee had a Personal Data Sheet, separate forms for leave applications, and two handwritten ledger cards. When employees needed to update their Personal Data Sheet, they had to visit the HR Office, retrieve their physical file, manually update the required information, and return the document for filing.

Leave applications followed a manual process in which a printed form was physically submitted to the Dean, HR Office, and Campus Director for approval. If the form was returned for correction, the employee had to revise and resubmit it. After



approval, HR personnel recorded the leave days on the handwritten ledger card, recalculated the remaining balance, and filed the documents. Employees also had to visit the HR Office to check their leave credits and submit requests, depending on HR personnel to locate records and calculate balances.

The reliance on physical records made documents vulnerable to loss, damage, and unauthorized access. The lack of a centralized digital repository also required HR personnel to spend additional time locating, reproducing, and verifying records. As the number of employees and administrative tasks increased, these manual procedures became more time-consuming and prone to errors, highlighting the need for a more organized and efficient digital approach to employee records and HR data management.

System Requirements of NexHRIS

The requirements identified during the requirements planning phase and validated with the HR Administrator were organized into functional, non-functional, hardware, and software requirements to define the system's features, performance, and technical needs.

Functional Requirements

Table 5 presents the functional requirements of NexHRIS, organized by module and describing the key functions and corresponding user roles authorized to access them.



Table 5. Functional Requirements of NexHRIS

Module	Functional Requirement	User Roles
Authentication	Sign in, email verification for privileged roles, account lockout, and password reset.	All roles
Employee Accounts	Create/update accounts, assign roles and organizational details, activate/deactivate accounts, and export directory.	HR Administrator
Form Templates	Upload, preview, and activate official forms.	HR Administrator
Personal Data Sheet	Download, submit, review, approve/return, and retain PDS versions.	Employee, Dean, Campus Director; HR Administrator (review/approval)
Leave Application	File, route, review, approve/return, resubmit, and print leave applications.	Employee, Dean, Campus Director, HR Admin
Ledger Cards	Post credits, leave, adjustments, and service credits; recalculate, view, and print cards.	HR Administrator; others for viewing
HR Policies	Publish, acknowledge, and track HR policies.	HR Administrator (publish); all other roles (read and acknowledge)
Digital ID Card	View, print, and publicly verify QR-based identification cards.	Employee, Dean, Campus Director



Non-Functional Requirements

Table 6 presents the non-functional requirements of NexHRIS and how the system satisfies each requirement. These requirements ensure that the system is secure, reliable, accessible, and easy to use for its intended users.

Table 2. Non-Functional Requirements of NexHRIS

Attribute	Requirement	Implementation in NexHRIS
Security and Privacy	Personnel data must be protected in accordance with the Data Privacy Act of 2012.	Role-based access, password hashing, two-factor verification, account lockout, private storage, sanitization, and HTTPS.
Accountability	Significant system actions must be traceable.	Activity logs record user actions, timestamps, IP addresses, and changes.
Document Fidelity	Official forms must retain their required format.	LibreOffice converts workbooks to A4 PDF.
Reliability	Approval routing and ledger balances must be accurate.	Role-based routing, automatic recalculation, and 530 automated tests.
Usability	The system must be easy to learn and use.	Role-specific menus, consistent design, and responsive layout.
Availability	The system must be accessible with an internet connection.	Web-based system hosted on a VPS.
Maintainability	The system must be maintainable by future developers.	Laravel MVC, database migrations, and versioned templates.



Hardware and Software Requirements

Table 7 presents the hardware and software requirements needed to operate NexHRIS on the server and client sides.

Table 3. Hardware and Software Requirements of NexHRIS

Category	Requirement
Server/Hosting	Hostinger KVM 1 VPS with 1 vCPU, 4 GB RAM, and 50 GB storage
Client Device	Desktop computer, laptop, tablet, or smartphone with an internet connection
Operating System	Ubuntu 24.04 LTS for the server; any operating system that supports a current web browser for client devices
Web Server and Runtime	Nginx and PHP 8.3
Database	MariaDB
Document Conversion	LibreOffice
Web Browser	Current version of Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari
Spreadsheet Application	Microsoft Excel (desktop) or LibreOffice Calc for completing official forms
Email	SMTP email service for verification codes, password resets, and notifications
Security	Let's Encrypt SSL/TLS certificate



System Design

The design of NexHRIS is documented in the appendices. The use case diagram in Appendix D presents the use cases of each role, named after the modules in the navigation menu of that role; the Dean and the Campus Director are drawn as specializations of the Employee, since they file their own leave and Personal Data Sheet through the same screens. The context diagram in Appendix E presents NexHRIS as a single process exchanging data with its four user roles, and the Level 1 data flow diagram in Appendix F decomposes it into nine processes and ten data stores, both in Gane and Sarson notation. The entity relationship diagram in Appendix G presents, in crow's foot notation, the eighteen tables of the relational database and their relationships. The flow chart in Appendix H presents the sign-in process and the modules available to each role, and the input-process-output chart in Appendix I summarizes the data that the system accepts, the processing it performs, and the outputs it produces.

Features of the Developed System

NexHRIS moves the HR transactions of the campus online while preserving the official Civil Service forms exactly as the campus uses them. The system is presented below by module; where a module behaves differently for each role, the corresponding screens are shown separately.

Authentication and Role-Based Access

Users sign in with their email address and password and must accept the Terms and Conditions of the system before signing in. Accounts that administer or approve on behalf of others the HR Administrator, the Deans, and the Campus Director are further



challenged with a six-digit verification code sent to their registered email address. The code expires after ten minutes, a new code may be requested after sixty seconds, and a code is invalidated after five incorrect entries. Employee accounts sign in with a password alone. After five consecutive failed sign-in attempts, an account is locked for fifteen minutes, and both the failed attempts and the lockout are written to the audit trail. A user who has forgotten the password may request a reset link by email.

After signing in, each user is taken to the dashboard of his or her role, and the navigation menu shows only the modules that the role may use. The four roles share one interface and differ only in what they may do. Their data boundaries are enforced in the database query rather than only in the interface, so that a Dean who edits a web address to reach the records of another college receives a refusal instead of data. All email messages sent by the system; verification codes, password reset links, and notifications follow one formal layout bearing the name of the institution.

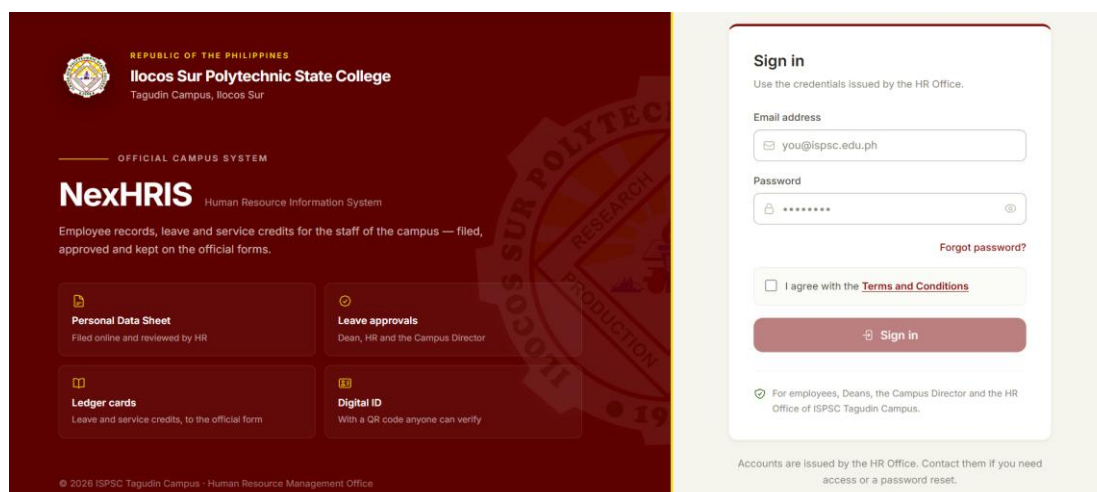


Plate 1. Sign-in Page with the Terms and Conditions

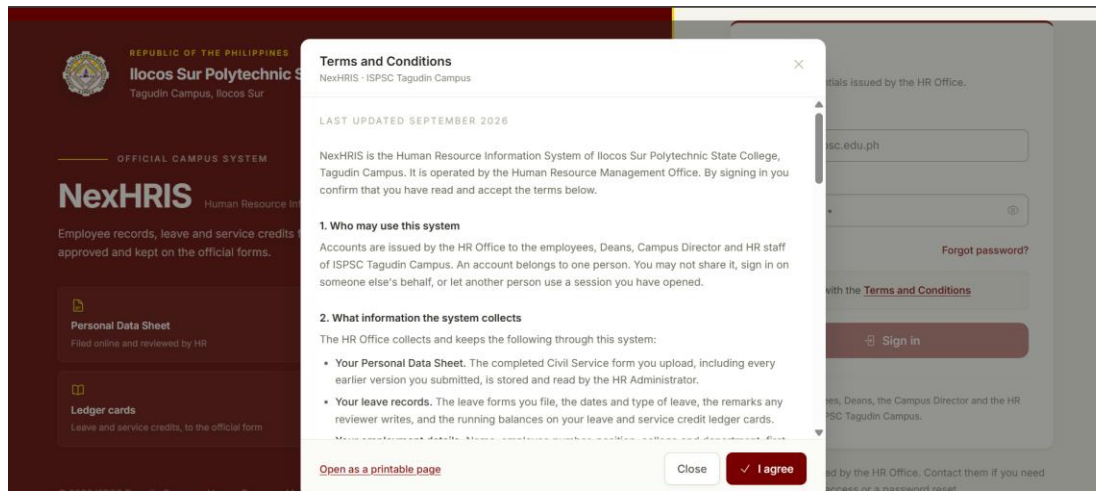


Plate 2. Terms and Conditions

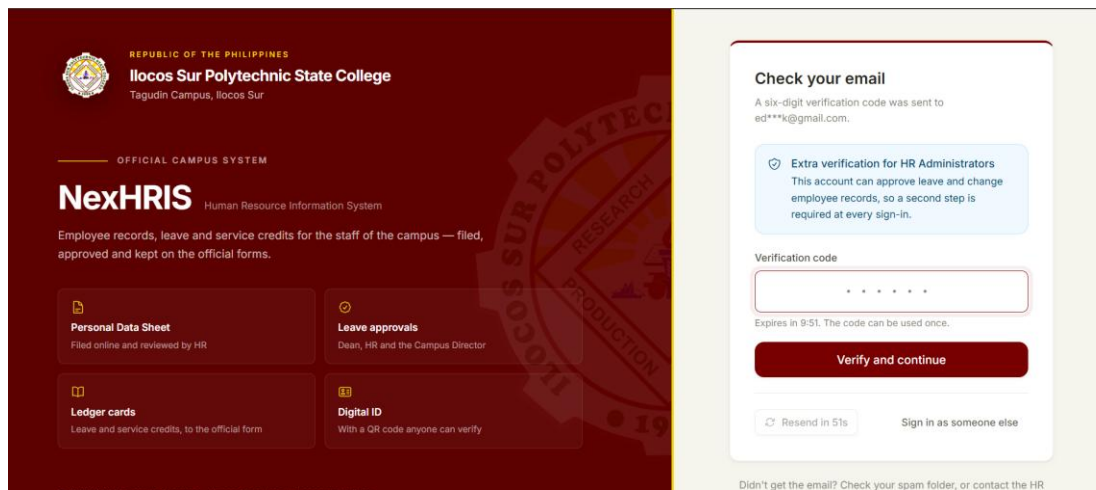


Plate 3. Two-Factor Verification

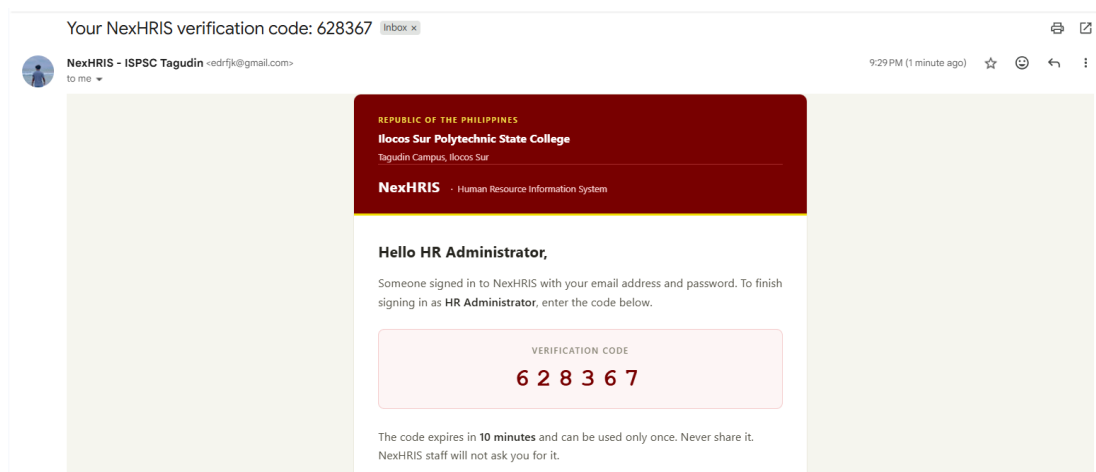


Plate 4. Verification Code Email



Plate 5. Password Reset Request

Plate 6. Refusal of Unauthorized Access

Dashboards

Each role opens to a different dashboard. The HR Administrator sees the number of employees, the leave applications in review, the approved applications awaiting ledger posting, and Personal Data Sheet compliance, together with a breakdown by college, the leave volume of the last six months, the distribution of leave types, the progress of the latest policy acknowledgment, newly added staff without a Personal Data Sheet, and recent activity. The Dean sees the headcount of his or her college, upcoming leave in the college, and his or her recent decisions. The Campus Director sees campus-wide



leave oversight, including the pending approvals by college, the leave volume of the last six months, the personnel on leave for the day, his or her own leave application, and recent decisions. The Employee sees his or her vacation leave, sick leave, and service credit balances, the leave used during the year, the status of his or her latest leave application and records, and the latest announcements.

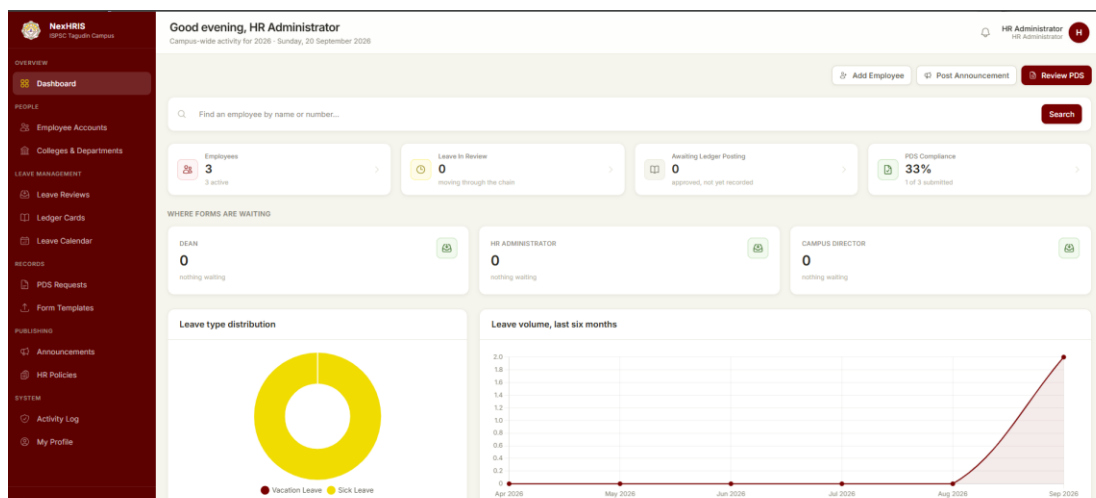


Plate 7. HR Administrator Dashboard

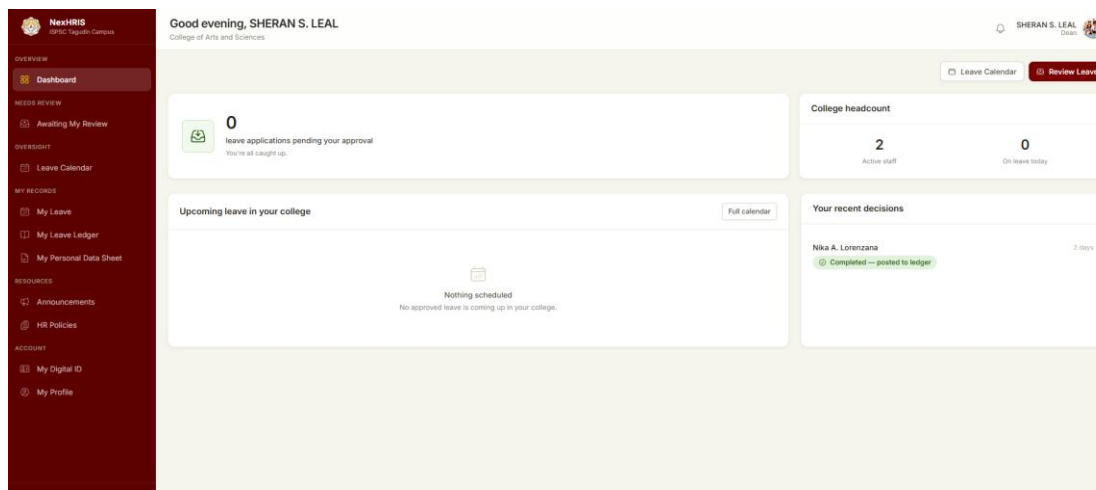


Plate 8. Dean Dashboard

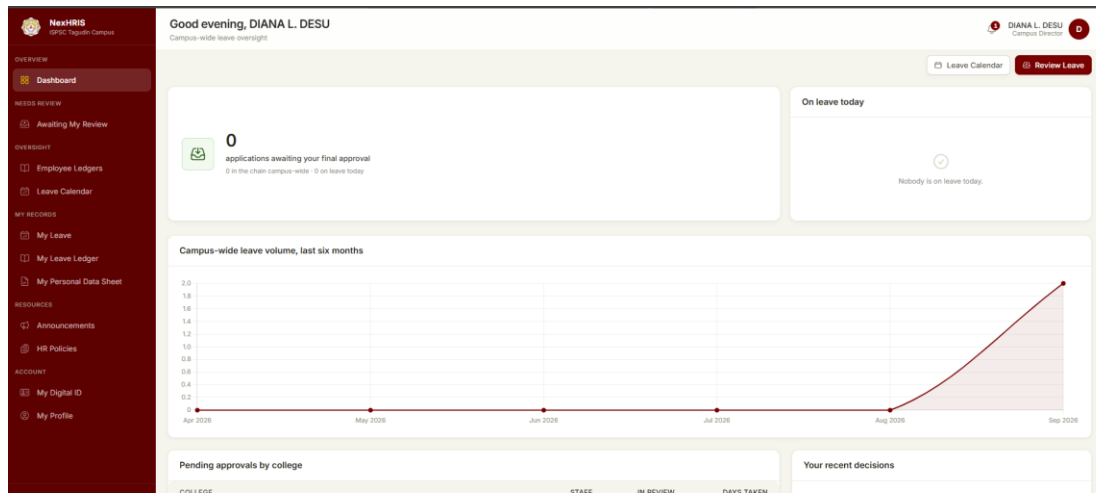


Plate 9. Campus Director Dashboard

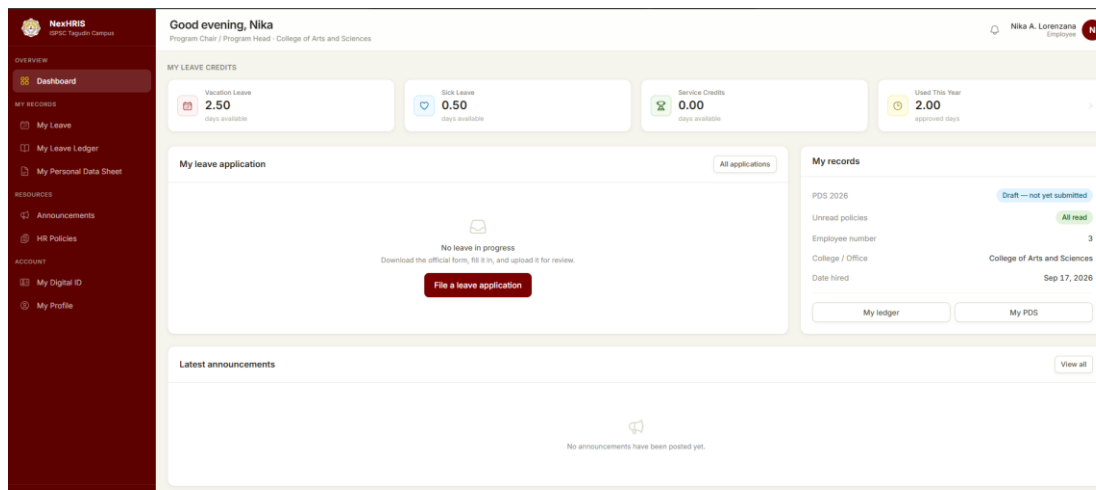


Plate 10. Employee Dashboard

Employee Accounts, Colleges, and Departments

The HR Administrator creates and maintains staff accounts, recording for each employee the employee number, position, role, college, department, first day of government service, and date hired. A department belongs to exactly one college, and the college to which an employee is assigned determines which Dean signs that employee's leave. Accounts are deactivated rather than deleted, so the history attached to a person is preserved after the person leaves the institution. The employee directory can be filtered and exported as a PDF file. From My Profile, every user maintains his



or her own name, email address, contact number, photograph, and password, while the employee number, position, college, department, and role remain under the control of the HR Administrator.

The screenshot shows the 'Employee Accounts' page in the NexHRIS system. The page has a sidebar with navigation options: OVERVIEW, PEOPLE, LEAVE MANAGEMENT, RECORDS, PUBLISHED, and SYSTEM. The main content area displays a summary of employee statistics: 2 Total Employees, 3 Active, 0 Inactive, and 3 New This Month. Below this is a search bar and filters for College / Office (CAS - College of Arts and Sciences) and Department (All departments). A table lists the employees:

EMPLOYEE	EMPLOYEE NO.	COLLEGE / PROGRAM	ROLE	STATUS	ACTIONS
Nika A. Lorenzana nika@gmail.com	3	CAS Bachelor of Arts in Psychology	—	Active	
SHERAN S. LEAL sheran.nstc24@gmail.com	1	CAS Bachelor of Science in Information Technology	Dean	Active	

Plate 11. Employee Accounts

The screenshot shows the 'Add Employee' form open in the NexHRIS system. The form is titled 'Add Employee' and includes instructions: 'Create the account and ledger-ready name in one place.' The form is divided into several sections:

- IDENTITY**: Fields for First Name, Middle Name, Last Name, Employee Number, and Email.
- EMPLOYMENT**: Fields for Position Category, Position, System Role, First day of government service, Contact Number, and College / Office.
- Department / Programme**: A dropdown menu to select the department.

Buttons for 'Cancel' and 'Create Employee' are at the bottom right of the form.

Plate 12. Creating an Employee Account



The screenshot shows the 'Employee Details' page for Nika A. Lorenzana, an Instructor I at the College of Arts and Sciences - BAP. The page includes a sidebar with navigation options like Dashboard, Employee Accounts, and Leave Management. The main content area displays a summary of the employee's status (Active), leave balances (VACATION LEAVE: 2.50, SICK LEAVE: 0.50, SERVICE CREDITS: 0.00), and a PDS 2028 draft. Below this, the 'Profile Details' section shows fields for Employee Number (3), First Name (Nika), Middle Name (A.), Last Name (Lorenzana), Email (nika@gmail.com), Contact Number (09281502315), Position Category (Academic Officials), Position (Instructor I), and System Role (Employee).

Plate 13. Employee Record

The screenshot shows the 'Colleges & Offices' page, which allows users to manage departments and their associated staff. It features a summary of departments (6), departments (24), and staff without a dean (5). A search bar is available to find departments by name or code. The page lists several departments under the 'CAS' (College of Arts and Sciences) category, including BACL (Bachelor of Arts in English Language), BAP (Bachelor of Arts in Psychology), BASS (Bachelor of Arts in Social Science), BIST (Bachelor of Science in Information Technology), and BSMT (Bachelor of Science in Mathematics). Each department entry shows the number of people, the department name, and options to edit or deactivate.

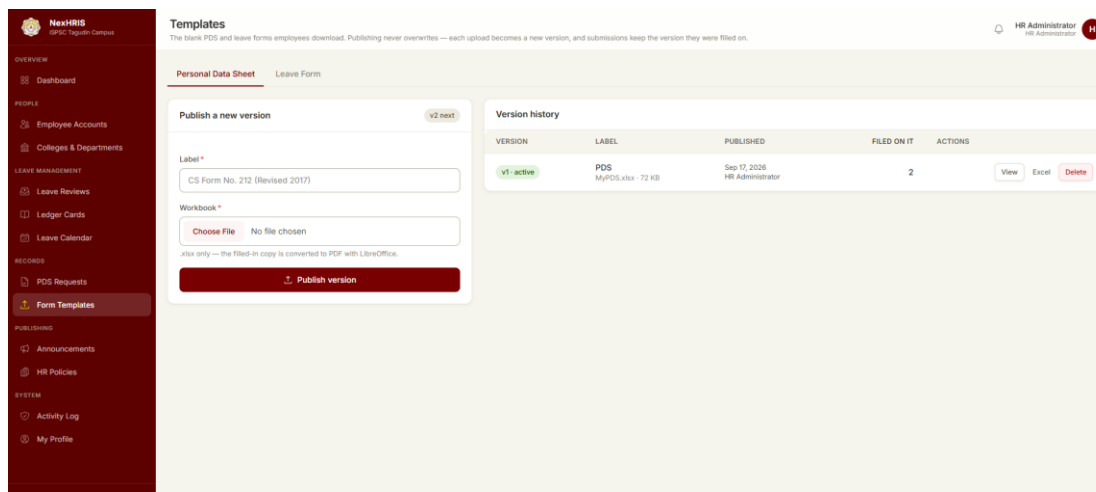
Plate 14. Colleges and Departments

The screenshot shows the 'My Profile' page for the HR Administrator. It includes a sidebar with navigation options like Dashboard, Employee Accounts, and Leave Management. The main content area displays the user's profile information, including a photo upload section (No file chosen), a 'Change photo' button, and a 'Choose File' button. The 'Account' section shows the system role (HR Administrator), signed in date (Sep 16, 2026), and a note about the administrator account. The 'Your details' section includes fields for Full name (HR Administrator), Email address (admin@ispcc.edu.ph), and Contact number (0900000000). The 'Change password' section includes fields for Current password, New password, and Confirm new password, with a 'Change password' button.

Plate 15. My Profile

Publication of the Official Forms

The official blank forms are held in the system rather than circulated by hand. From Form Templates, the HR Administrator uploads the Application for Leave (CS Form No. 6) and the leave ledger card, while the Personal Data Sheet (CS Form No. 212) is published from PDS Requests. Each upload becomes a new numbered version, and activating a version retires the previous one instead of overwriting it, so that every submission remains associated with the version on which it was filled. A published form can be previewed as a PDF before it is issued to staff, and the original workbook remains available for download.



VERSION	LABEL	PUBLISHED	FILED ON IT	ACTIONS
v1-active	PDS MyPDS.xlsx - 72 KB	Sep 17, 2026 HR Administrator	2	View Excel Delete

Plate 16. Form Templates

Personal Data Sheet

From My Personal Data Sheet, an employee downloads the active CS Form No. 212, completes it in Microsoft Excel or LibreOffice Calc, and uploads the completed workbook. The system accepts only a workbook based on the official template, and the employee may view the converted PDF before submitting it to the HR Office. The HR Administrator reviews each submission from PDS Requests, reads it as a PDF in the browser or downloads the original workbook, and either approves it or returns it with



remarks. A returned sheet displays the remarks to the employee, who uploads a corrected version and submits it again, and every earlier version is retained. The employee is notified of each decision by email and through the notification bell. Deans and the Campus Director submit their own Personal Data Sheets through the same screens. Because the exported PDF is a conversion of the workbook that the employee actually uploaded rather than a form rebuilt from stored values, the printed sheet is the official form of the campus, named after the employee and the year it covers.

Personal Data Sheet
Download the official template, fill it out, then upload it back here.

Step 1 — Download the Official Template

Step 2 — Upload Your Completed PDS

Your PDS is already uploaded — see the summary above. Click "Upload a Different Version" if you need to make changes.

Plate 17. My Personal Data Sheet

PDS Requests
Monitor employee Personal Data Sheet submissions.

Summary:

- Not Started: 1 (33%)
- Draft: 1 (33%)
- Submitted: 1 (33%)
- Approved: 0 (0%)
- Returned: 0 (0%)

Table:

EMPLOYEE	COLLEGE / DEPARTMENT	STATUS (2026)	SUBMITTED ON	REVIEW
DIANA L. DESU	Non-Teaching Staffs	Not started	—	ⓘ
Nika A. Lorenzana	College of Arts and Sciences	Draft	—	ⓘ
SHERAN S. LEAL	College of Arts and Sciences	Submitted	Sep 17, 2026 7:48 PM	ⓘ

Plate 18. PDS Requests



Plate 19. Reviewing a Personal Data Sheet

Plate 20. Converted Personal Data Sheet

Leave Application and Approval Chain

From My Leave, an employee views his or her leave balances, downloads the active CS Form No. 6, and completes it. The employee then files the application by selecting vacation leave or sick leave, entering the inclusive dates and the reason, and uploading the completed form as a workbook, a Word document, or a PDF file. The application is routed according to the rank of the applicant, since no user signs his or her own leave: the application of an employee passes through the Dean of the employee's college, the HR Administrator, and the Campus Director; the application of a Dean skips the Dean stage; and the application of the



Campus Director is decided by the HR Administrator alone.

Each reviewer finds the applications awaiting his or her stage in a review queue—Awaiting My Review for the Dean and the Campus Director, and Leave Reviews for the HR Administrator—reads the form as a PDF in the browser, and either approves it or returns it with remarks. A Dean sees only the applications of his or her own college. The next reviewer is notified by email and through the notification bell. A returned application displays the remarks to the employee, who uploads a corrected form, and the corrected application restarts from the first stage of the chain. The employee follows the application through a stage-by-stage timeline and, once it is fully approved, prints the approval sheet, on which any skipped stage is marked not applicable.

Plate 21. Filing a Leave Application



My leave applications All statuses

Sick Sep 17, 2026 – Sep 18, 2026 2 day(s)
testing

Completed — posted to ledger

✓ Dean SHERAN S. LEAL · Sep 17

✓ HR Administrator HR Administrator · Sep 17

✓ Campus Director DIANA L. DESU · Sep 17

[View as PDF](#) [My upload](#) [Print approval sheet](#)

Plate 22. Tracking a Leave Application

NexHRIS ISPC Tagudin Campus

Leave Reviews
Forms waiting on your signature as Dean

APPROVAL CHAIN
Dean · you · HR Administrator · Campus Director

Employees upload their filled-in leave form here first. Reviewing it online means nobody prints and chases signatures for a form that was going to be sent back. You see only employees registered under your program.

Search Employee name or number

Leave type All leave types

Order Longest waiting first

Oldest first is the order an approval queue is worked in.

[Apply filters](#)

Waiting on you 2 forms

EMPLOYEE	LEAVE	INCLUSIVE DATES	DAYS	WAITING	PROGRESS	ACTION
Nika A. Lorenzana 3 College of Arts and Sciences - Bachelor of Arts in Psychology	Vacation	Sep 29, 2026 – Sep 30, 2026	2	Today	1 Dean 2 HR Administrator 3 Campus Director	Review
Nika A. Lorenzana 3 College of Arts and Sciences - Bachelor of Arts in Psychology	Vacation	Oct 1, 2026 – Oct 2, 2026	2	Today	1 Dean 2 HR Administrator 3 Campus Director	Review

Recently reviewed by you

Nika A. Lorenzana
Sick leave · Sep 17, 2026

Completed — posted to ledger [View](#)

Plate 23. Dean Review Queue

NexHRIS ISPC Tagudin Campus

Leave Form - Nika A. Lorenzana
Reference LV-00002

STATUS
Awaiting HR review

APPROVAL CHAIN
Dean SHERAN S. LEAL · Sep 20

HR Administrator Waiting

Campus Director

Leave details

Type of leave
Vacation Leave

Working days
2.00

Inclusive dates
September 20, 2026 – September 30, 2026

Submitted
September 20, 2026 10:22 PM

Reason given
test

Uploaded leave form

Official-LV-Form (3) (1).xlsx

[View as PDF](#) [Original](#)

Employee

Nika A. Lorenzana

Position
Instructor I

College / Office
College of Arts and Sciences

Department
Bachelor of Arts in Psychology

VL 2.50 SL 0.50 SERVICE 0.00

Credits cover this: 2.00 of 2.50 vacation leave credits.

View full ledger card →

Your decision
As HR Administrator

[Approve](#) [Return](#)

Plate 24. HR Administrator Leave Review



NexHRIS
Ilocos Sur Polytechnic State College

Leave Form - Nika A. Lorenzana
Reference LV-00002

STATUS
Awaiting Campus Director review

Leave details
Type of leave: Vacation Leave
Working days: 2.00
Inclusive dates: September 20, 2026 - September 30, 2026
Submitted: September 20, 2026 10:22 PM
Reason given: test

Employee
Nika A. Lorenzana
Position: Instructor I
College / Office: College of Arts and Sciences
Department: Bachelor of Arts in Psychology

Uploaded leave form
Official-Leave-Form (3) (1).xlsx
View as PDF Original

Your decision
As Campus Director
Approve Return

Plate 25. Campus Director Final Approval

NexHRIS
Ilocos Sur Polytechnic State College

Leave Form - Nika A. Lorenzana
Reference LV-00002

Leave details
Type of leave: Vacation Leave
Working days: 2.00
Inclusive dates: September 20, 2026 - September 30, 2026
Submitted: September 20, 2026 10:22 PM
Reason given: test

Employee
Nika A. Lorenzana
Position: Instructor I
College / Office: College of Arts and Sciences
Department: Bachelor of Arts in Psychology

Uploaded leave form
Official-Leave-Form (3) (1).xlsx
View as PDF Original

Your decision
As HR Administrator
What needs correcting? *
e.g. The inclusive dates do not match the attached medical certificate.
Return to employee Cancel

Plate 26. Returning an Application with Remarks

My leave applications
1 in review All statuses

Vacation Sep 21, 2026 - Sep 22, 2026 2 day(s)
test

Leave details
Type of leave: Vacation
Working days: 2.00
Inclusive dates: Sep 21, 2026 - Sep 22, 2026
Submitted: Sep 21, 2026 10:22 PM
Reason given: test

Employee
Nika A. Lorenzana
Position: Instructor I
College / Office: College of Arts and Sciences
Department: Bachelor of Arts in Psychology

Uploaded leave form
Official-Leave-Form (3) (1).xlsx
View as PDF Original

Your decision
As HR Administrator
View as PDF My upload

Plate 27. Dean's Own Approval Sheet



Leave Calendar

The leave calendar shows the approved and pending leave of a chosen month, scoped to the role viewing it. The Dean sees only his or her own college, while the HR Administrator and the Campus Director see the whole campus and may narrow the view to one college. The calendar can be filtered by leave type, and the month displayed can be exported as a PDF file for posting.

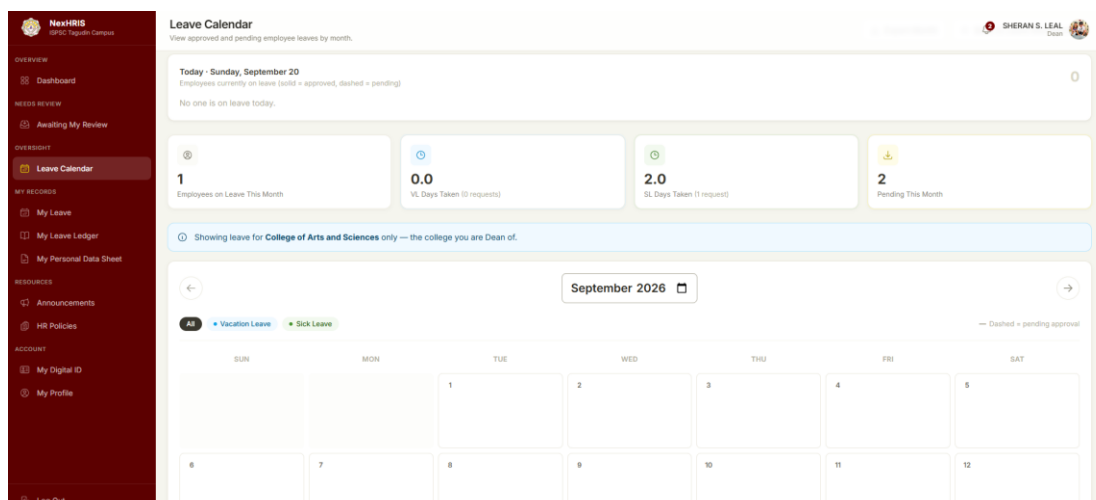


Plate 28. Leave Calendar of a College

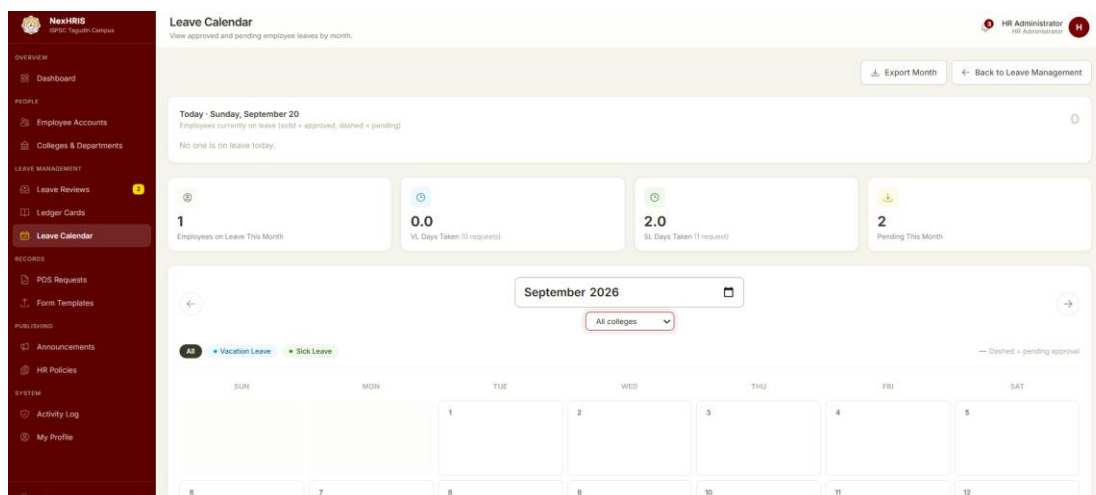


Plate 29. Campus-Wide Leave Calendar



Leave and Service Credit Ledger Cards

The campus keeps two ledger cards per employee on the same official form. The leave ledger card records the vacation and sick leave credits earned and used and the resulting balances, while the service credit ledger card records credits earned outside regular accrual, such as for election duty, enrolment, or accreditation work, and the days charged against them. Once an application is fully approved, the HR Administrator posts it to the ledger and decides on which card the days are charged. The HR Administrator also posts monthly leave credits to one employee or to all personnel in a single action, in which a period already credited is skipped rather than credited twice, and records adjustments.

Each line stores the balances as they stood after that line, so correcting or removing a line recalculates every balance below it. The leave card is kept to two decimal places and the service credit card to three, as the forms of the campus require, and every correction is written to the audit trail together with the values that the line previously held. The HR Administrator maintains all cards, the Campus Director views the card of every employee through Employee Ledgers, and each employee reads his or her own through My Leave Ledger. The cards print on A4 paper in the layout of the official form.



Plate 30. Ledger Cards

Plate 31. Maintaining a Ledger Card

Plate 32. Posting Approved Leave to the Ledger



My Leave Ledger
Your official ledger card, as maintained by HR. Read-only — ask HR if a figure looks wrong.

VACATION LEAVE: 0.50 days available

SICK LEAVE: 0.50 days available

SERVICE CREDITS: 0.00 days available

Official ledger card

EMPLOYEE'S LEAVE LEDGER CARD

NAME: LORENZANA, NIK A. OFFICE: College of Arts and Sciences

FIRST DAY OF GOVERNMENT SERVICE: September 17, 2026

PERIOD		REMARKS	VACATION LEAVE			SICK LEAVE		
FROM	TO		EARNED	ABSENCE / UNDERTIME / W/PAY	BALANCE	EARNED	ABSENCE / UNDERTIME / W/PAY	BALANCE
Sep 1, 2026	Sep 30, 2026	Earned 2026-09-01 - 2026-09-30	1.25		1.25	1.25		1.25
Sep 17, 2026	Sep 18, 2026	SL — testing			2.5		2	0.5
Sep 29, 2026	Sep 30, 2026	VL — test		2	0.5			0.5
Oct 3, 2026	Oct 5, 2026	Earned 2026-10-03 - 2026-10-05	1.25		2.5	1.25		2.5
BALANCE AS OF SEPTEMBER 20, 2026					0.50			0.50

Plate 33. My Leave Ledger

EMPLOYEE'S LEAVE LEDGER CARD

NAME: LORENZANA, NIK A. OFFICE: College of Arts and Sciences

FIRST DAY OF GOVERNMENT SERVICE: September 17, 2026

PERIOD		REMARKS	VACATION LEAVE			SICK LEAVE		
FROM	TO		EARNED	ABSENCE / UNDERTIME / W/PAY	BALANCE	EARNED	ABSENCE / UNDERTIME / W/PAY	BALANCE
Sep 1, 2026	Sep 30, 2026	Earned 2026-09-01 - 2026-09-30	1.25		1.25	1.25		1.25
Sep 17, 2026	Sep 18, 2026	SL — testing			2.5		2	0.5
Sep 29, 2026	Sep 30, 2026	VL — test		2	0.5			0.5
Oct 3, 2026	Oct 5, 2026	Earned 2026-10-03 - 2026-10-05	1.25		2.5	1.25		2.5
BALANCE AS OF SEPTEMBER 20, 2026					0.50			0.50

Generated September 20, 2026 10:59 PM

HRSPSC Tagudin Campus - Human Resource Management Office

Plate 34. Printed Ledger Cards

Management Reports

In addition to the documents of individual employees, the system produces campus-wide reports: the leave balances of all personnel as a PDF file or an Excel workbook, the employee directory as a PDF file that respects the filters on the screen, and the leave calendar of a chosen month. Every generated document is named after the person or the period it covers, so that a saved file can be identified without opening it.



EMPLOYEE NO.	NAME	COLLEGE / OFFICE	VL BALANCE	SL BALANCE	SERVICE CREDITS
2	DIANA L. DESU	ADMIN	0.00	0.00	0.00
3	Nika A. Lorenzana	CAS	0.50	0.50	0.00
1	SHERAN S. LEAL	CAS	0.00	0.00	0.00

Plate 35. Leave Balances Report

Nika A. Lorenzana's Leave Ledger Card.pdf

1 / 2

|

100%

+

1

2

EMPLOYEE'S LEAVE LEDGER CARD

NAME: LORENZANA

NIKA

A.

OFFICE: College of Arts and Sciences

(FAMILY NAME)

(FIRST NAME)

(M.I.)

FIRST DAY OF GOVERNMENT SERVICE: September 17, 2026

PERIOD			VACATION LEAVE				SICK LEAVE			
FROM	TO	REMARKS	EARNED	ABSENCE / UNDERTIME / W/PAY	ABSENCE / UNDERTIME / W/O PAY	BALANCE	EARNED	ABSENCE / UNDERTIME / W/PAY	ABSENCE / UNDERTIME / W/O PAY	BALANCE
Sep 1, 2026	Sep 30, 2026	Earned 2026-09-01 – 2026-09-30	1.25			1.25	1.25			1.25
Sep 17, 2026	Sep 18, 2026	SL – testing				2.5		2		0.5
Sep 29, 2026	Sep 30, 2026	VL – test		2		0.5				0.5
Oct 3, 2026	Oct 5, 2026	Earned 2026-10-03 – 2026-10-05	1.25			2.5	1.25			2.5
BALANCE AS OF SEPTEMBER 20, 2026						0.50				0.50

Generated September 20, 2026 11:04 PM

ISPEC Taguin Campus - Human Resource Management Office

Plate 36. Employee Directory Report

HR Policies

The HR Administrator publishes policies as formatted text written in a rich-text editor, as an uploaded file, or as a link, with an effective date and, optionally, an expiry date. A policy may be marked as requiring acknowledgment, pinned to the top of the list, and published or withdrawn. Policy text is sanitized before it is displayed, so that no script embedded in a policy can run in the browser of a reader. Employees, Deans, and the Campus Director read the published policies and acknowledge those that require it, and the system records when each person viewed and acknowledged a



policy. The HR Administrator sees who has and who has not acknowledged each policy, and the progress of the latest policy is shown on the dashboard.

Plate 37. HR Policies Management

Plate 38. Creating a Policy



EMPLOYEE	VIEWED	ACKNOWLEDGED
DIANA L. DESU 2	—	Pending
Nika A. Lorenzana 2	—	Pending
SHERAN S. LEAL 1	—	Pending

Plate 39. Policy Acknowledgment Tracking

EMPLOYEE	VIEWED	ACKNOWLEDGED
DIANA L. DESU 2	—	Pending
Nika A. Lorenzana 2	—	Pending
SHERAN S. LEAL 1	—	Pending

Plate 40. Reading and Acknowledging a Policy

Announcements and Notifications

The HR Administrator posts announcements to the whole campus or to the personnel of one college. Users are notified of new announcements, of each stage of their leave applications, and of decisions on their Personal Data Sheets by email and through the notification bell, and the Notifications page lists every notice with the option to mark it as read.



NexHRIS
Ilocos Sur Polytechnic State College

Announcements
Post notices to the whole campus or a single college.

Test Announcements has been posted.

[New Announcement](#)

TITLE	AUDIENCE	POSTED	STATUS	ACTIONS
Test Announcements Testing Announcements	Campus-wide	Sep 20, 2026 HR Administrator	Published Pinned	Delete

Plate 41. Announcements Management

NexHRIS
Ilocos Sur Polytechnic State College

Announcements
Notices from the HR Office.

[Pinned](#)

Test Announcements
Testing Announcements

Posted by HR Administrator

Sep 20, 2026

Plate 42. Announcements

Notifications [Mark all read](#)

- Test Announcements**
Testing Announcements
4 seconds ago
- Approved leave awaiting a ledger posting**
Nika A. Lorenzana's leave is fully approved. Record the days and credits used on their ledger card.
17 minutes ago
- A leave form needs your review**
SHERAN S. LEAL has submitted a leave form awaiting your approval as HR Administrator.
28 minutes ago

[New Announcement](#)

[Delete](#)

Plate 43. Notifications



Digital Identification Card

Employees, Deans, and the Campus Director are each issued a digital identification card showing the photograph, name, position, college, and employee number of the holder, together with a QR code. The card can be printed, and the QR code resolves to a public verification page that confirms the name, position, college, and account status of the holder, so that a card presented outside the campus can be checked without an account and without revealing any other personal information.

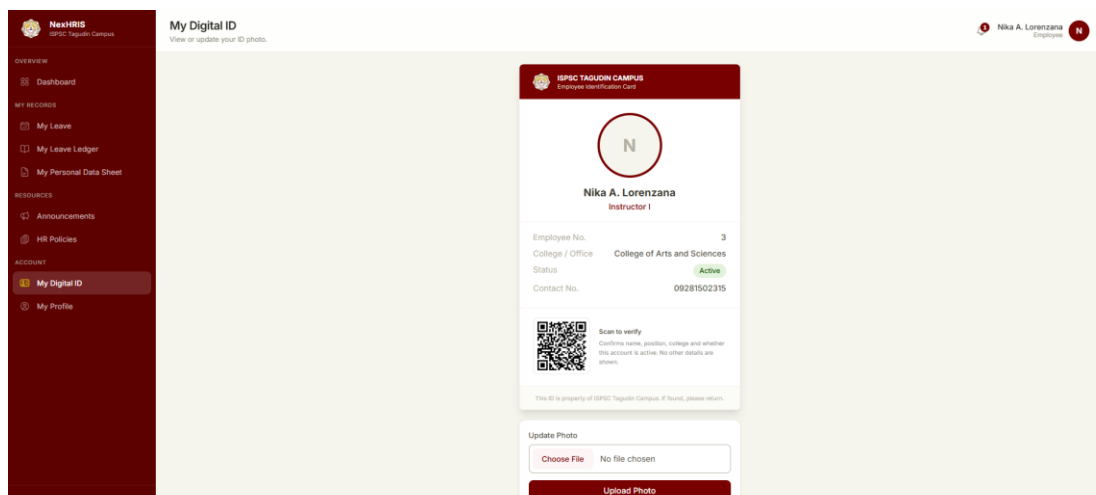


Plate 44. My Digital ID

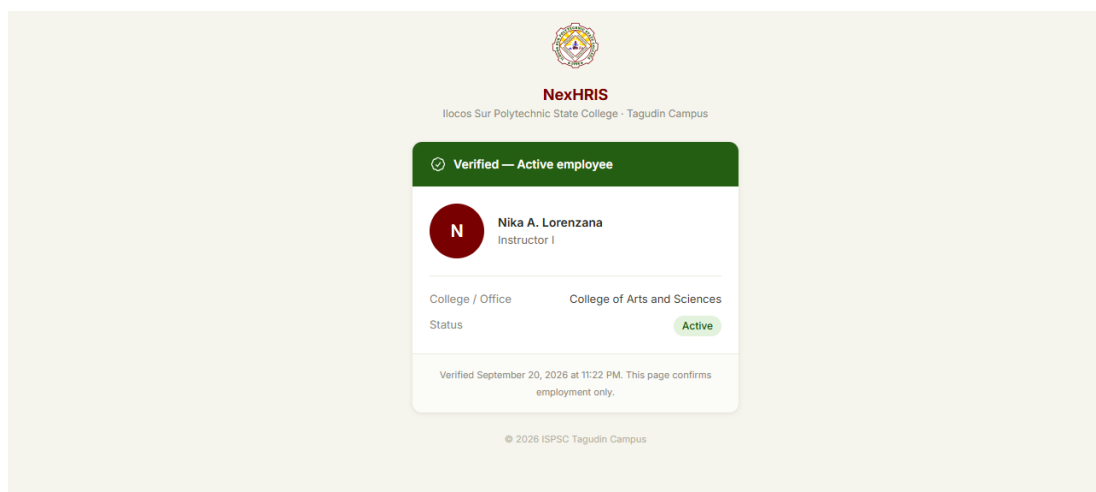


Plate 45. Public Verification Page



Activity Log

Every sign-in, failed sign-in, logout, decision on a leave application or Personal Data Sheet, publication of a form template or policy, change to an account, and correction of a ledger card is recorded together with the account, date and time, IP address, and browser behind it. The HR Administrator searches the log by keyword, action, user, and date range, giving the HR Office a reliable trail for accountability.

WHEN	WHO	ACTION	DETAILS	RECORD	IP ADDRESS
Sep 20, 2026 11:24:09 PM	HR Administrator HR Administrator	Auth login	HR Administrator signed in.	User #1	124.217.80.128
Sep 20, 2026 11:23:51 PM	Nika A. Lorenzana Employee	Auth logout	Nika A. Lorenzana signed out.	User #4	124.217.80.128
Sep 20, 2026 11:23:46 PM	Nika A. Lorenzana Employee	Auth login	Nika A. Lorenzana signed in.	User #4	124.217.80.128
Sep 20, 2026 11:23:49 PM	HR Administrator HR Administrator	Auth logout	HR Administrator signed out.	User #1	124.217.80.128
Sep 20, 2026 11:13:26 PM	HR Administrator HR Administrator	Announcement posted	Posted announcement "Test Announcements".	Announcement #1	124.217.80.128
Sep 20, 2026	HR Administrator	Auth login	HR Administrator signed in.	User #1	124.217.80.128

Plate 46. Activity Log

Data Security

To protect personnel data, NexHRIS applies role-based access control that distinguishes the permissions of the HR Administrator, the Campus Director, the Dean, and the Employee, and enforces these boundaries at the query level. Uploaded personnel records—completed Personal Data Sheets, filed leave forms, and policy attachments—are held outside the web-accessible directory and are released only through the application, so each request is checked against the role of the person making it; only the blank official forms, which contain no personal data, and profile photographs are served directly. Passwords are stored hashed and must contain at least eight characters, including both letters and numbers, and every connection to the



system is encrypted over HTTPS. With these measures, NexHRIS enhances the security, accessibility, and efficiency of employee records management at ISPSC Tagudin Campus.

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APPENDICES



APPENDIX B

COMMUNICATION LETTER

ILOCOS SUR POLYTECHNIC STATE COLLEGE
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Page 1 of 2

College of Arts and Sciences

VISION
UIP 2030: A smart, globally competitive and responsive state university.

MISSION
UIP is committed to producing globally competent and virtuous human resources, generating advanced knowledge, and developing innovative technologies for the sustainable development of society.

CORE VALUES
Innovativeness
Motivation
Professionalism
Excellence
Transparency and Accountability
Unity and Teamwork
Sustainability

MOTTO
Sapientia et Virtus in Posterum (Wisdom and Virtue for the Future)

DEVELOPMENT GOALS
Onwards UIP
Optimize institutional quality and reputation.
Nurture excellence in academics and services.
Widen research, development, and innovation programs.
Advance responsive extension services and vibrant external linkages.
Revitalize capacity for resource management and generation.
Develop capability and competence of human resources.
Sustain ethical leadership and good governance.
Underscore advocacies for sustainable development.
Intensify internationalization initiatives.
Promote modernization of physical and digital infrastructure.

June 1, 2026

DANIEL JUAN B. RAMIREZ, PhD.
Campus Director
Attention: **MRS. RHEA B. GACUTAN**
HRMO II

Sir:

Warm greetings and goodwill!

We, the undersigned Bachelor of Science in Information Technology (BSIT) students of Ilocos Sur Polytechnic State College, respectfully request permission to conduct our capstone project entitled "NexHRIS: A Smart and Next-Generation Human Resource Information System for Ilocos Sur Polytechnic State College" within the campus, particularly in the Human Resource (HR) Office and other related offices.

This project aims to develop a web-based HR system that will help improve the management and storage of employee records and documents for academic and institutional use only.

We sincerely hope for your approval and support. Thank you for your time and consideration.

Respectfully yours,

EIDREF JAKE S. MANALANSAN
Researcher
YURI BIE P. SOMERA
Researcher

DIANA SOFIA L. LOREZNANA
Researcher
SHERAN S. LEAL
Researcher

ARVIE KEITH MANZANO
Researcher

Noted by:
JIM-MAR F. DELOS REYES, MIT
BSIT Program Head

DR. JOY G. BEA, DIT
OIC Dean, CAS / Adviser

Approved:
DANIEL JUAN B. RAMIREZ, PhD.
Campus Director

UWRI
The World's Leading University Rankings by WURI
Times Higher Education Impact Rankings 2025
ISO 9001:2015
CERTIFIED QUALITY MANAGEMENT SYSTEM
IAS

APPENDIX C

User Case Diagram

Nexhris: A Smart and Next-Generation Human Resource Information System for
Ilocos Sur Polytechnic State College

